

H Society of SEC

MVP Architecture Document – Hybrid AI Security Monitoring System

This document outlines the Minimum Viable Product (MVP) for a hybrid AI-driven network monitoring system designed for corporate environments. The system focuses on balancing security, performance, and client trust by using a hybrid architecture that processes data both locally and centrally.

Problem Statement

Organizations require continuous monitoring of their networks to detect anomalies, threats, and inefficiencies. However, traditional solutions often require sending large volumes of sensitive data to centralized systems, raising concerns about privacy, bandwidth, and trust.

Proposed Solution (Hybrid Model)

The MVP uses a hybrid architecture where lightweight client agents perform local data collection and preprocessing, while a centralized server performs aggregation, analysis, and visualization. This approach minimizes data transfer while maintaining intelligence.

System Architecture Overview

- 1 Client Agents: Deployed within corporate networks to collect system and network data.
- 2 Local Processing: Filters noise, aggregates metrics, and detects basic anomalies.
- 3 Secure Communication Layer: Encrypts all outgoing data using TLS and authenticates agents.
- 4 Central Server: Receives summarized data, stores events, and performs higher-level analysis.
- 5 Dashboard Service: Displays insights, alerts, and network statistics to clients.

Data Strategy

- 1 Raw logs remain primarily on the client system.
- 2 Only summarized metrics and critical events are transmitted to the server.
- 3 Sensitive data is never transmitted without explicit permission.
- 4 Data retention policies limit storage duration (e.g., 7 days local, 30 days server).

Security Considerations

- 1 End-to-end encryption (TLS) for all communications.
- 2 Agent authentication using API keys or tokens.
- 3 Verification of agent identity to prevent spoofing.
- 4 Secure update mechanism for client agents.

- 5 Configurable data-sharing policies for clients.

MVP Scope

- 1 Basic client agent collecting system and network metrics.
- 2 Backend API for receiving and storing summarized data.
- 3 Simple dashboard for visualization of metrics and alerts.
- 4 Rule-based anomaly detection (no advanced AI initially).

Future Enhancements

- 1 Advanced anomaly detection using machine learning.
- 2 Cross-client intelligence and pattern recognition.
- 3 Automated threat response capabilities.
- 4 Localization of models for region-specific network behavior (e.g., African infrastructure).

This MVP provides a scalable and secure foundation for building a next-generation network monitoring system. By prioritizing trust, efficiency, and modular design, H Society of SEC can evolve the system into a powerful cybersecurity platform.